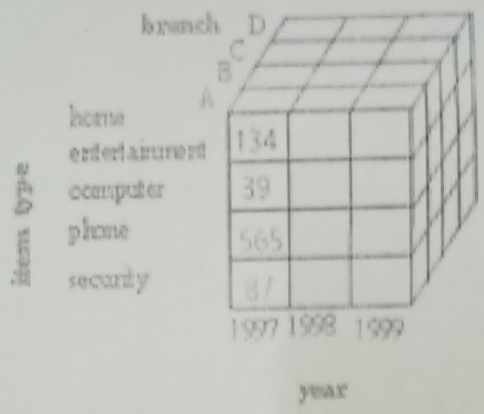
Vidyalankar Institute of Technology

Semester V – INFT - Mid Semester Assessment – I

Date: 05/08/2024	Data Warehouse and Mining	20 Marks/ 45 Minutes

			CO
1	**Solve any two (5 marks each).**		
	A	Define Data Warehouse, which are the properties of Data Warehouse with example.	CO1
	B	Explain the KDD process in detail.	CO4
	C	Explain the application of data mining in healthcare. Discuss how data mining techniques can be utilized to predict diseases, improve patient care, and optimize hospital resource management. Provide specific examples to illustrate your points.	CO1
2	**Solve any two (5 marks each).**		
	A	Define a factless fact table and provide a suitable example to illustrate its use in a data warehouse. Explain the scenarios where a factless fact table is useful and how it differs from a traditional fact table.	CO1
	B	Differentiate between OLTP and OLAP.	CO1
	C	Dimensional Modelling is preferred over E-R modelling for data Warehouse, Justify.	CO2
2	**Solve any one (10 marks each).**		
	A	Explain OLAP operations and perform these operations on the Vidyalankar retail company that operates in different branches and sells a diverse range of products. The company has implemented an OLAP system to analyze its sales data for strategic decision-making.	CO2



i) Perform slice operation for home-entertainment.
ii) Perform Dice operation for item of home entertainment and computer in the year of 1997 and 1998
iii) Perform drill down operation for the product of computer as PC, Laptop, Printer

	B	Consider a data warehouse for a hospital that includes the following dimensions: Doctor, Patient, and Time. The warehouse also contains two measures: Count and Fee. 1. Draw a star schema and a snowflake schema for this data warehouse. 2. Explain the differences between a star schema and a snowflake schema.	CO2

CO1	Demonstrate a comprehensive understanding of the fundamental concepts of data warehousing and its architecture.
CO2	Design a data warehouse schema using dimensional modelling techniques and explain the ETL process involved in data warehousing.
CO3	To introduce concepts and fundamentals of data lakes
CO4	Understand data mining principles and perform data preprocessing and Visualization.
CO5	Understand the concept of data mining and identify appropriate data mining algorithms to solve real-world problems.
CO6	Implement basic data mining algorithms such as classification, clustering, and association mining.

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Question-1 (a)

Data Warehouse:- It is a centralized repository designed to store and manage large volumes of structured data from multiple sources, enables complex querying and analysis for decision-making and reporting.

Properties of Data warehouse:-

① **Subject Oriented:-** Data warehouse is organized around major subjects of the business like sales, finance etc.
e.g. A retail data warehouse may have separate subject areas for sales transactions, customer ~~identifi~~ information, inventory levels and suppliers data.

② **Integrated:-** Data from various sources is integrated into a coherent format within the warehouse.
e.g. An industry may integrate the data of CRM system from various resources.

③ **Time-Variant:-** Data warehouses stores historical data, allowing users to analyze trends over time.

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		e.g. A financial data warehouse might store monthly sales figure for several years.
	(4)	Non Volatile :- Once data is entered into a data warehouse, it remains unchanged. e.g. In a healthcare data warehouse, patients records are not modified after they are entered.

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Question-1 (b)

Knowledge Discovery in Databases (KDD):-

→ KDD process is a comprehensive approach to extract valuable info. and insights from large datasets.

→ It involves several stages:-

① **Data Selection** is initial stage involves identifying and selecting relevant data from various sources.

② **Data Preprocessing** is required once the data is selected. Data gets cleaned and transformation is performed thereafter.

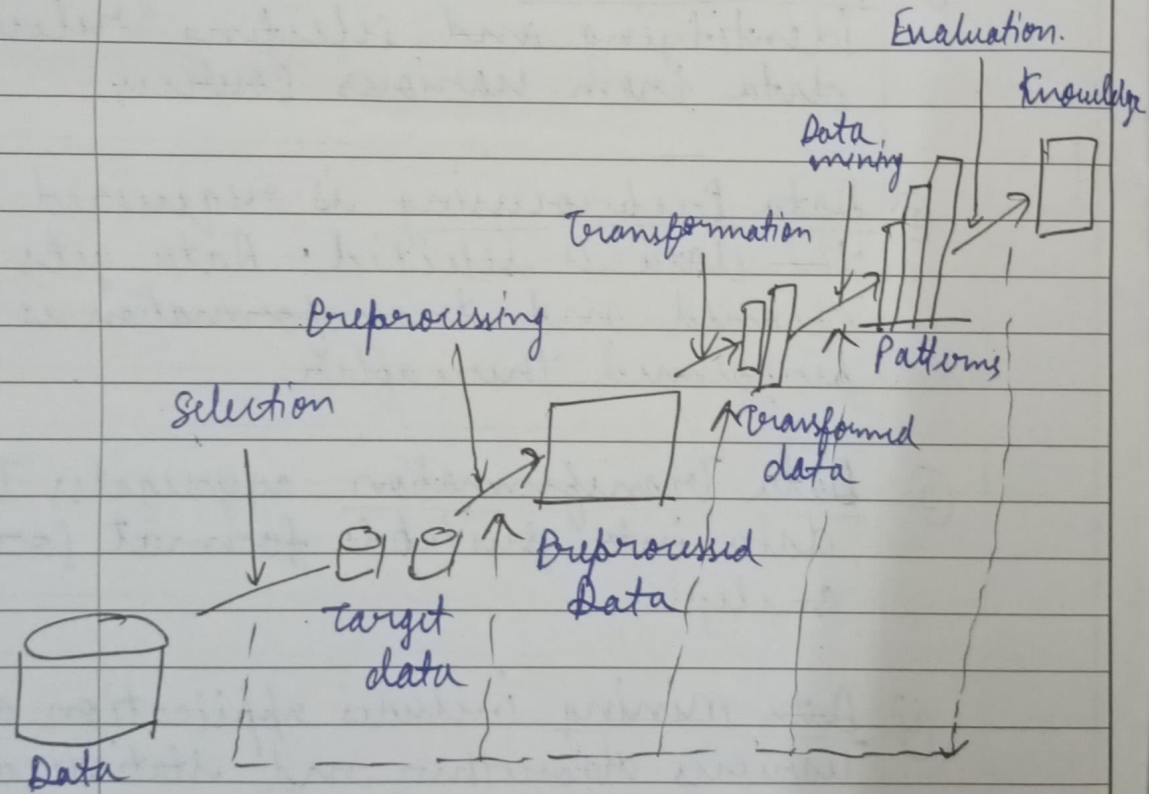
③ **Data Transformation** aggregates the data into suitable format for analysis.

④ **Data Mining** includes application of various algorithm and statistical methods to discover pattern in data.

⑤ **Evaluation** checks the validity and usefulness of mined data.

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	(6)	Knowledge Representation stage presents the gained insights in comprehensible manner.



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Question-1(c)

Data Mining in Healthcare :-

Data Mining in healthcare involves analyzing large volume of medical data to uncover patterns and relationships that can improve various aspects of healthcare delivery.

Following are the areas where data mining can help :-

① **Predicting diseases :-**

→ Various classification techniques can help segregating patients based on their risk factor.

→ ~~This~~ is Regression analysis can help identifying relationship b/w risk factors and disease prevalence.

② **Improving Patient Care :-**

→ Clustering algorithms can group patients with similar health conditions.

→ Association mining can identify relation b/w diff. variables such as symptoms and diagnose.

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	(3)	Optimizing Hospital Resource Mgmt:-
	→	Time series analysis used to predict future trends in patient admissions and resource needs.

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Question-2 (A)

Factless Fact table :-

It is a type of fact table in data warehouse that does not contain measurable facts or numerical values.

Example : Student_Enrollment.

Dimensions :
- → Student_ID
- → Cours_ID
- → Semester_ID

Now the table will look like :-

Student_ID	Course_ID	Semester_ID
1001	2001	2023-Fall
1002	2002	2023-Spring
1001	2003	2023-Spring.

Usefulness :-

- → Event tracking.
- → Event Analysis.

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Difference from Traditional Fact Table:-

→ Traditional table contains quantitative data whereas Fact table does not.

→ Traditional table focus on summarization and aggregating numerical measures, whereas factless fact table focus on the occurrence of events and relationship b/w dimensions.

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Question-2(B)

Difference b/w OLTP and OLAP:-

	OLTP (Online Transaction Processing)	OLAP (Online Analytical Processing)
(1)	Transaction oriented	Subject-oriented.
(2)	High CRUD activity	High read activity.
(3)	Real time information	Historical Information
(4)	Operational database.	Informational Database.
(5)	Highly volatile data.	Non-volatile data.
(6)	Data redundancy is bad.	Data redundancy handled well.
(7)	Few level of granularity.	Multiple level of granularity.
(8)	RDBMS	RDBMS / MDBMS

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Question - 2 (c)

Dimensional modelling preffered over ER modeling :-

→ It is because of its advantages in query performance, ease of use and alignment with business processes.

Justification :-

① **Optimized Query Performance :-**

Dimensional modelling (DM) structures data into star or snowflake schemas which are designed to simplify and speed up complex queries and aggregations. Whereas E-R models require complex joins and can slow down query performance.

② **User friendly design :-**

Dimensional models are intuitive for business users and analysts because they reflect business processes and metrics in a straightforward manner, which is not the case with E-R modelling.

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	(3)	Alignment with Business Process :-

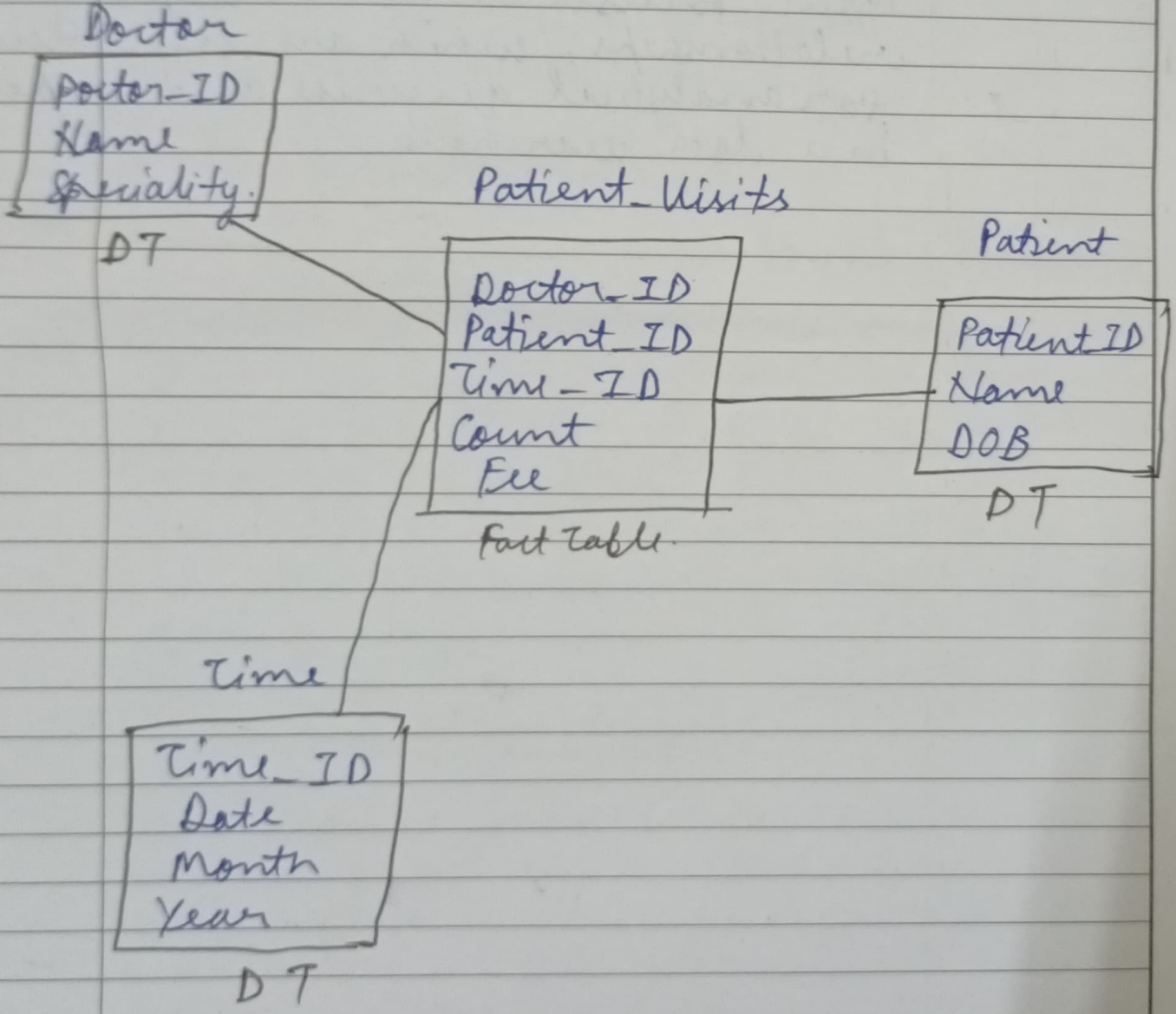
Dimensional modeling is designed to align closely with how business data is analyzed and reported. E-R modeling, while suitable for transactional systems, often focuses on data integrity and relationships, which are less relevant for analytical queries and reporting in a data warehouse.

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Question - 3 (B)

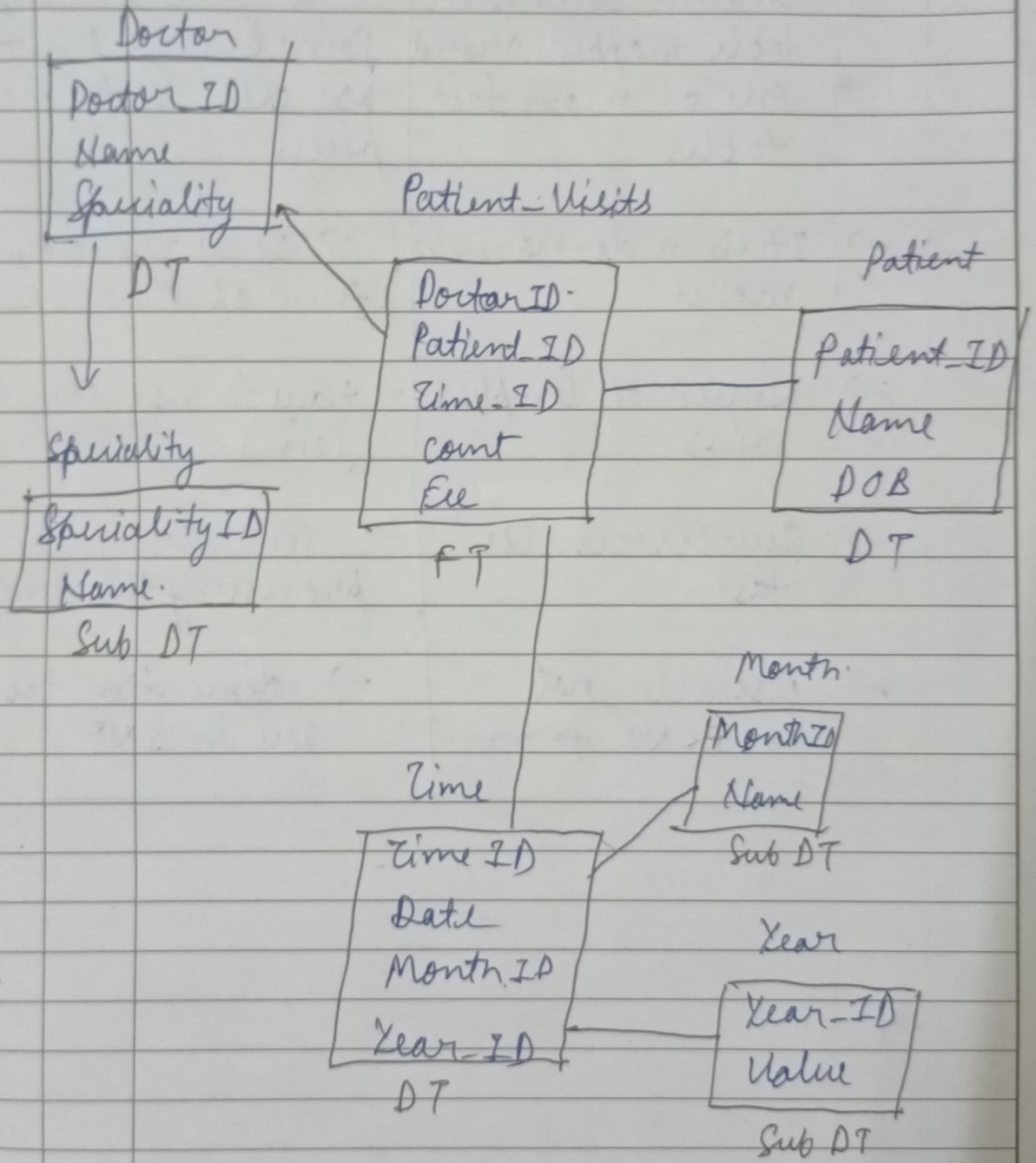
Star Schema:- Star Schema of data warehouse for a hospital with dimensions Doctor, Patient and Time will look like:-



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Snowflake Schema :-



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	Star Schema	Snowflake Schema
→	Contains dimension table mapped around one or more fact tables.	→ Contains in-depth joins because the tables are split in to many pieces.
→	It is a de-normalized model.	→ It is the normalized form of Star Schema.
→	Noneed of Complicated joins.	→ Have to use complicated joins.
→	Queriesresults fast.	→ Some delay in processing the Query.
→	usually not in BCNF form	→ dimention tables are in 3NF.

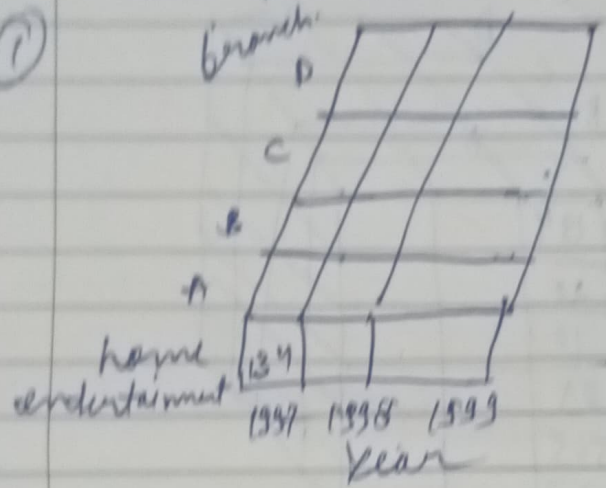
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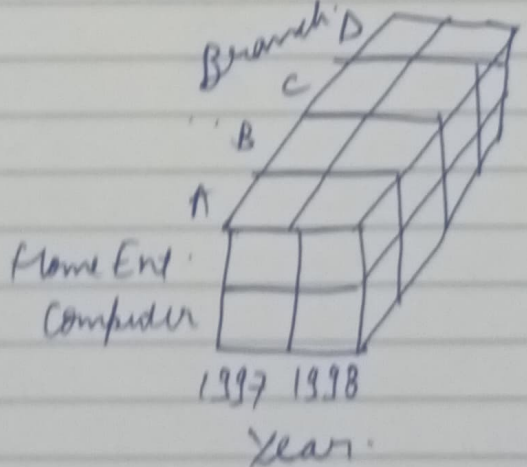
Question - 3 (A)

Slice operation :-

(i)



(a) **Slice Operation :-**



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	(3)	Drill Down :-

